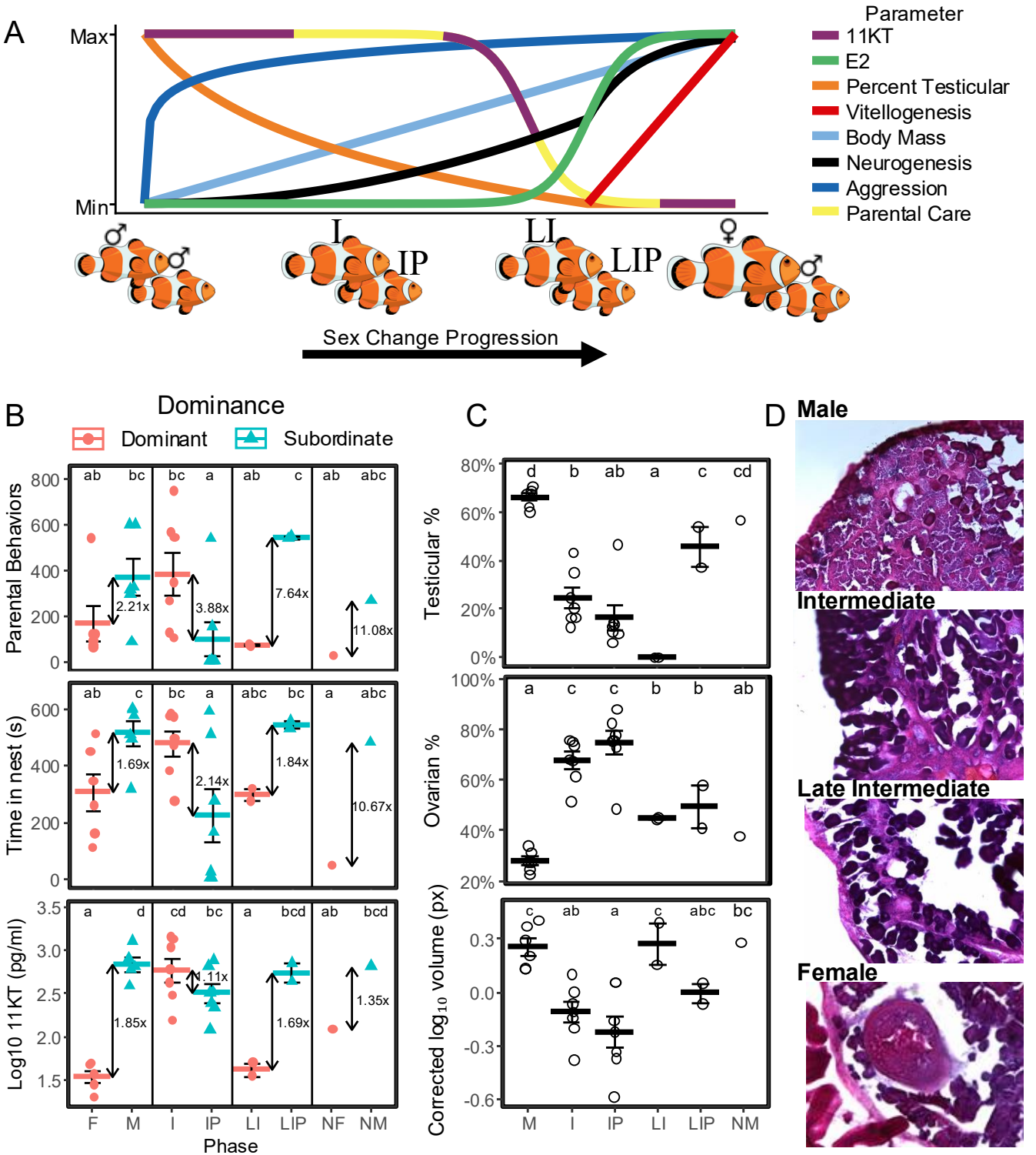

A
Max
Min
Parameter
11KT
E2
Percent Testicular
Vitellogenesis
Body Mass
Neurogenesis
Aggression
Parental Care
I
IP
LI
LIP
Sex Change Progression
B
Dominance
Dominant
Subordinate
Parental Behaviors
Time in nest (s)
Log10 11KT (pg/ml)
2.21x
3.88x
7.64x
11.08x
1.69x
2.14x
1.84x
10.67x
1.85x
1.11x
1.69x
1.35x
F
M
I
IP
LI
LIP
NF
NM
Phase
C
Testicular %
Ovarian %
Corrected log10 volume (px)
M
I
IP
LI
LIP
NM
D
Male
Intermediate
Late Intermediate
Female

Fig.1. Male, intermediate, and female anemonefish display distinct physiology and behavior.
(A) Schematic outlining the behavioral and physiological changes that occur during protandrous sex change in anemonefish.
(B) Top panel: Parental behaviors observed during a parental care assay. Middle panel: Time spent in the nest during parental care assay; a proxy for parental care. Bottom Panel: Log10 transformed serum 11-ketotestosterone.
(C)Top panel: Percent of the gonad made of testicular tissue. Middle panel: Percent of the gonad made of ovarian tissue. Bottom panel: Gonadal volume corrected for the individual's length (methods).
(D) Example hematoxylin and eosin stains of gonads from the major phases of sex change. In B, plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.Arrows indicate fold-difference between the two groups. In B, C, crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different.

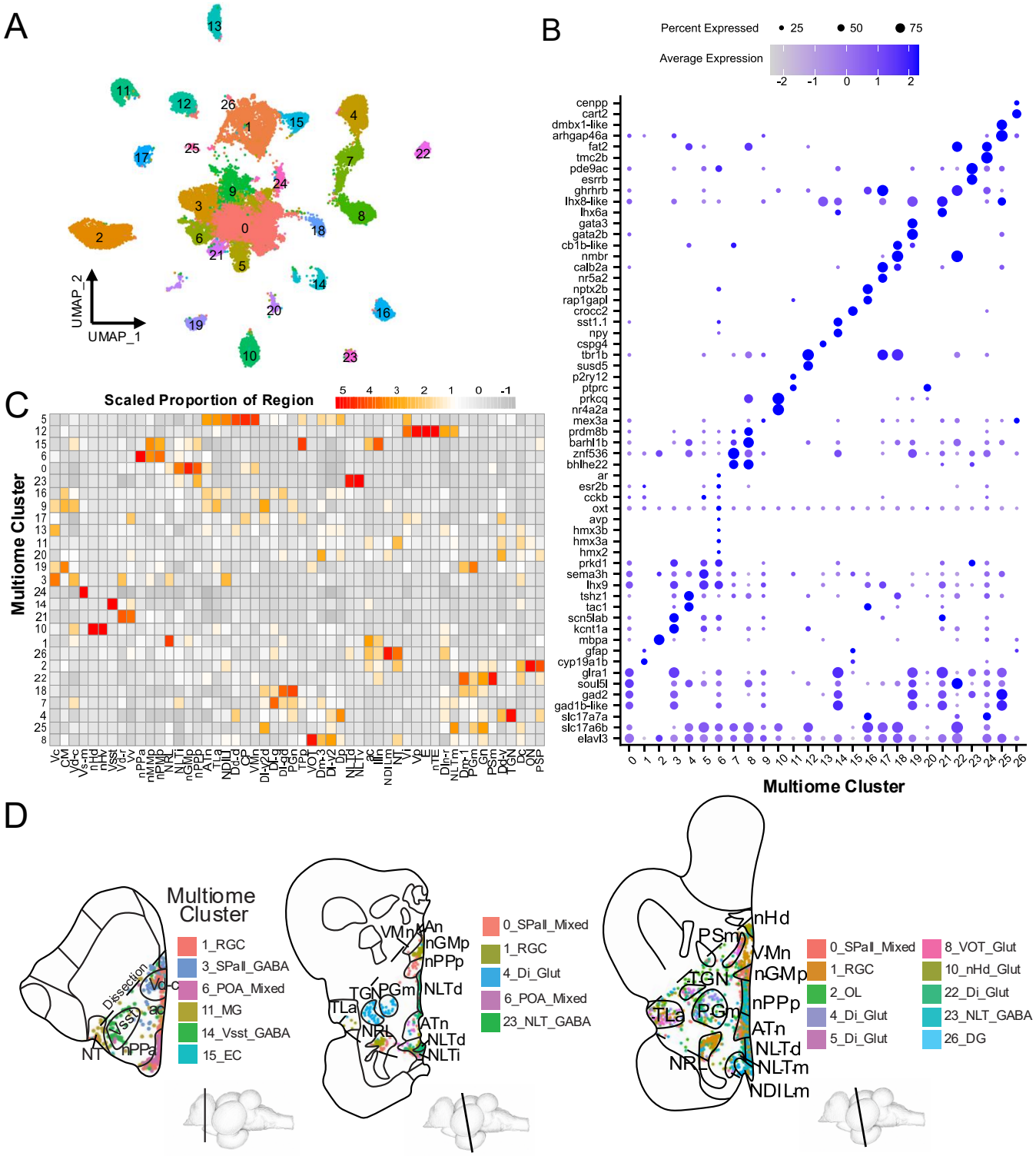

A
UMAP_2
UMAP_1
B
Percent Expressed
25
50
75
Average Expression
-2
-1
0
1
2
Multiome Cluster
C
Scaled Proportion of Region
5
4
3
2
1
0
-1
Multiome Cluster
D
Multiome Cluster
1_RGC
3_SPall_GABA
6_POA_Mixed
11_MG
14_Vsst_GABA
15_EC
Dissection
Vc-c
ac
Vsst
NT
nPPa
VMn
MAn
nGMp
nPPp
TGN
PGm
NLTd
TLa
ATn
NRL
NLTi
0_SPall_Mixed
1_RGC
4_Di_Glut
6_POA_Mixed
23_NLT_GABA
nHd
PSm
VMn
nGMp
TGN
nPPp
TLa
PGm
ATn
NRL
NLTd
NLTm
NDILm
0_SPall_Mixed
1_RGC
2_OL
4_Di_Glut
5_Di_Glut
8_VOT_Glut
10_nHd_Glut
22_Di_Glut
23_NLT_GABA
26_DG

Fig.2. Multiomics capture high quality transcriptomes, epigenomes, and spatial transcirptomes,
(A) UMAP embeddings of our sn-Multiome data with each cluster numbered.
(B) Dotplot showing canonical and data-driven markers for each of our multiome clusters.
(C) Heatmap showing the scaled proportion of each neuroanatomical region made of each multiome cluster (see supplement xx for abbreviations).
(D) Example projections of multiome clusters onto representative brain sections. The dashed line indicates approximately where we dissected brains for sn-multiomics analysis. Below each section is a sketch of an *A. ocellaris* brain with a line indicating the approximate sectioning plane for the given section.

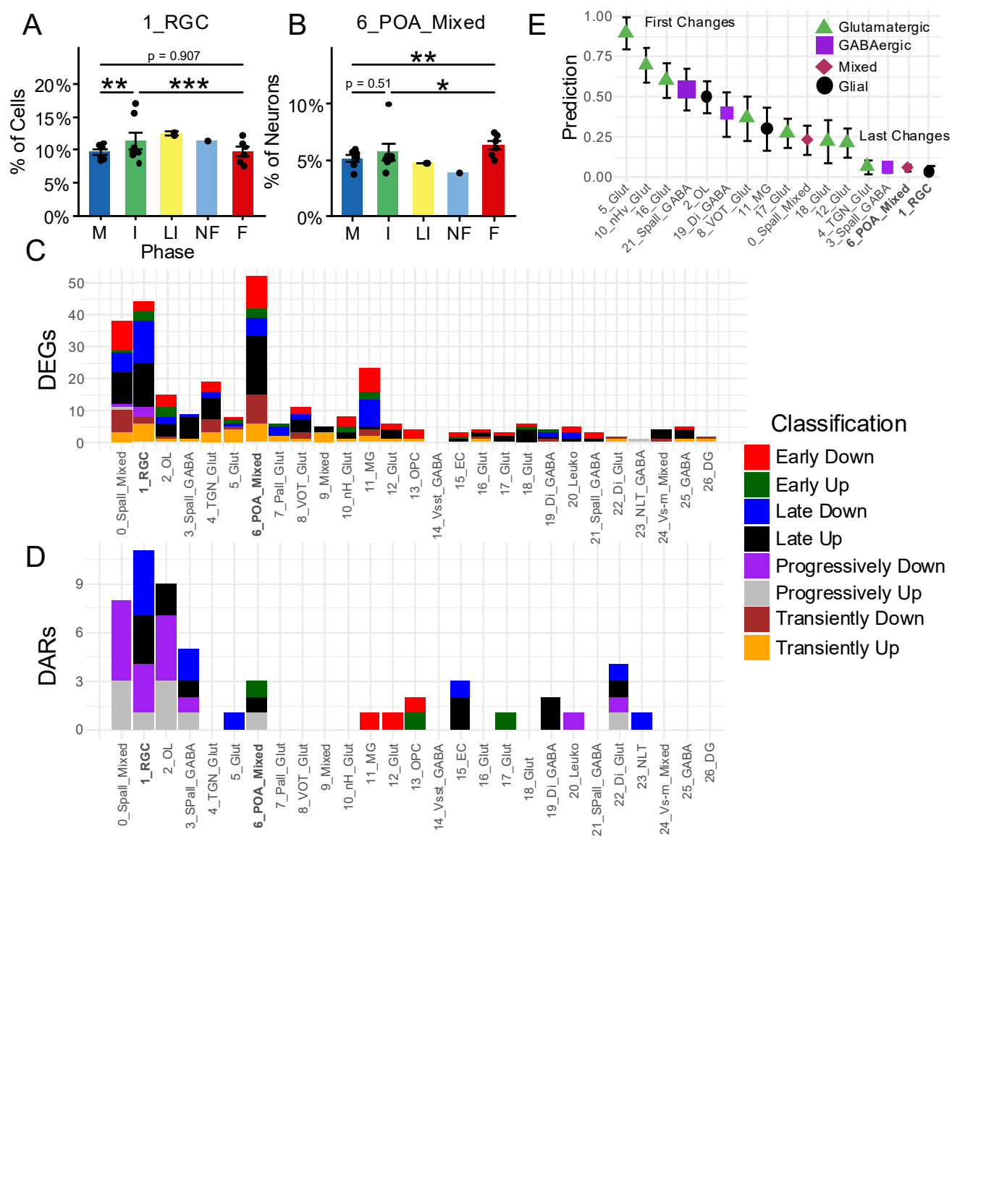

A
1_RGC
p = 0.907
**
***
% of Cells
20%
15%
10%
5%
0%
M
I
LI
NF
F
Phase
B
6_POA_Mixed
**
p = 0.51
*
% of Neurons
10%
5%
0%
M
I
LI
NF
F
E
First Changes
Last Changes
Glutamatergic
GABAergic
Mixed
Glial
Prediction
1.00
0.75
0.50
0.25
0.00
5_Glut
10_nHv_Glut
16_Glut
21_Spall_GABA
2_OL
19_Di_GABA
8_VOT_Glut
11_MG
0_Spall_Mixed
17_Glut
18_Glut
12_Glut
4_TGN_Glut
3_Spall_GABA
6_POA_Mixed
1_RGC
C
DEGs
50
40
30
20
10
0
0_Spall_Mixed
1_RGC
2_OL
3_Spall_GABA
4_TGN_Glut
5_Glut
6_POA_Mixed
7_Pall_Glut
8_VOT_Glut
9_Mixed
10_nH_Glut
11_MG
12_Glut
13_OPC
14_Vsst_GABA
15_EC
16_Glut
17_Glut
18_Glut
19_Di_GABA
20_Leuko
21_Spall_GABA
22_Di_Glut
23_NLT_GABA
24_Vs-m_Mixed
25_GABA
26_DG
Classification
Early Down
Early Up
Late Down
Late Up
Progressively Down
Progressively Up
Transiently Down
Transiently Up
D
DARs
9
6
3
0
0_Spall_Mixed
1_RGC
2_OL
3_SPall_GABA
4_TGN_Glut
5_Glut
6_POA_Mixed
7_Pall_Glut
8_VOT_Glut
9_Mixed
10_nH_Glut
11_MG
12_Glut
13_OPC
14_Vsst_GABA
15_EC
16_Glut
17_Glut
18_Glut
19_Di_GABA
20_Leuko
21_SPall_GABA
22_Di_Glut
23_NLT
24_Vs-m_Mixed
25_GABA
26_DG

Fig.3. Cells change composition, transcriptome, and epigenome asynchronously during sex change.
(A) The percent of all cells belonging to 1_RGC.
(B) The percent of all neurons belonging to 6_POA_Mixed.
(C) The number of differentially expressed genes (DEGs) by multiome cluster.
(D) The number of differentially accessible regions (DARs) by multiome cluster.
(E) Machine learning predictions for the mean sex of intermediates based on their expression of DEGs and DARs by multiome cluster; a proxy for the temporal order of sex change Shape and color represents broad cell type.
In C, D, bars are colored based on our classification for the manner in which they change sex (see methods). In A, B, points indicate the proportion for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

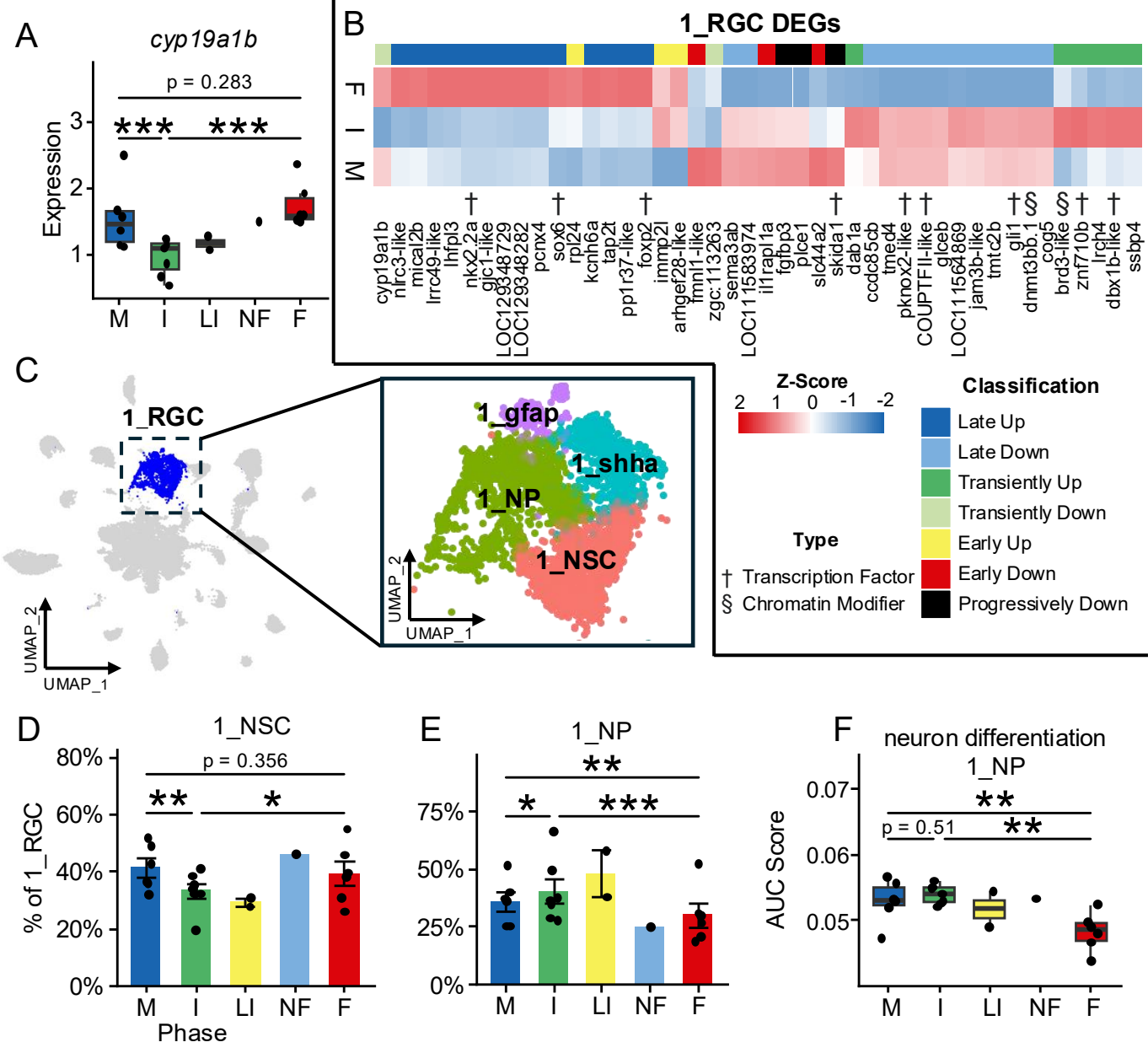

A
cyp19a1b
p = 0.283
***
***
Expression
M
I
LI
NF
F
B
1_RGC DEGs
F
I
M
cyp19a1b
nfrc3-like
mical2b
lrrc49-like
lhfpl3
nkx2.2a
gjc1-like
LOC129348729
LOC129348282
pcnx4
sox6
rpl24
kcnh6a
tap2t
pp1r37-like
foxp2
immp2l
arhgef28-like
fmnl1-like
zgc:113263
sema3ab
LOC111583974
il1rapl1a
fgfbp3
plce1
slc44a2
skida1
dab1a
ccdc85cb
tmed4
pknox2-like
COUPTFII-like
glceb
LOC111564869
jam3b-like
tmtc2b
gli1
dnmt3bb.1
cog5
brd3-like
znf710b
lrrc4
dbx1b-like
ssbp4
C
1_RGC
1_gfap
1_shha
1_NP
1_NSC
UMAP_2
UMAP_1
Z-Score
2
1
0
-1
-2
Classification
Late Up
Late Down
Transiently Up
Transiently Down
Early Up
Early Down
Progressively Down
Type
† Transcription Factor
§ Chromatin Modifier
D
1_NSC
p = 0.356
**
*
% of 1_RGC
80%
60%
40%
20%
0%
M
I
LI
NF
F
Phase
E
1_NP
**
*
***
75%
50%
25%
0%
M
I
LI
NF
F
F
neuron differentiation
1_NP
**
p = 0.51
**
AUC Score
0.07
0.06
0.05
M
I
LI
NF
F

Fig.4. Radial glial cells undergo changes in transcriptome and subtype composition over the course of sex change.
(A) Brain aromatase (*cyp19a1b*) expression in 1_RGC.
(B) A heatmap showing all DEGs by phase in 1_RGC. Cells are colored by z-score of the mean expression for each phase. The top bar indicates the classification for how each gene changes sex, and the symbols below the matrix indicate what kind of gene each DEG is.
(C) UMAP showing subclustering of 1_RGC.
(D) Percent of 1_RGC composed of 1_NSC.
(E) Percent of 1_RGC composed of 1_NP.
(F) AUCell score of a gene expression module composed of genes labeled with the GO term “neuron differentiation” in 1_NP.
In A, D-F, points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

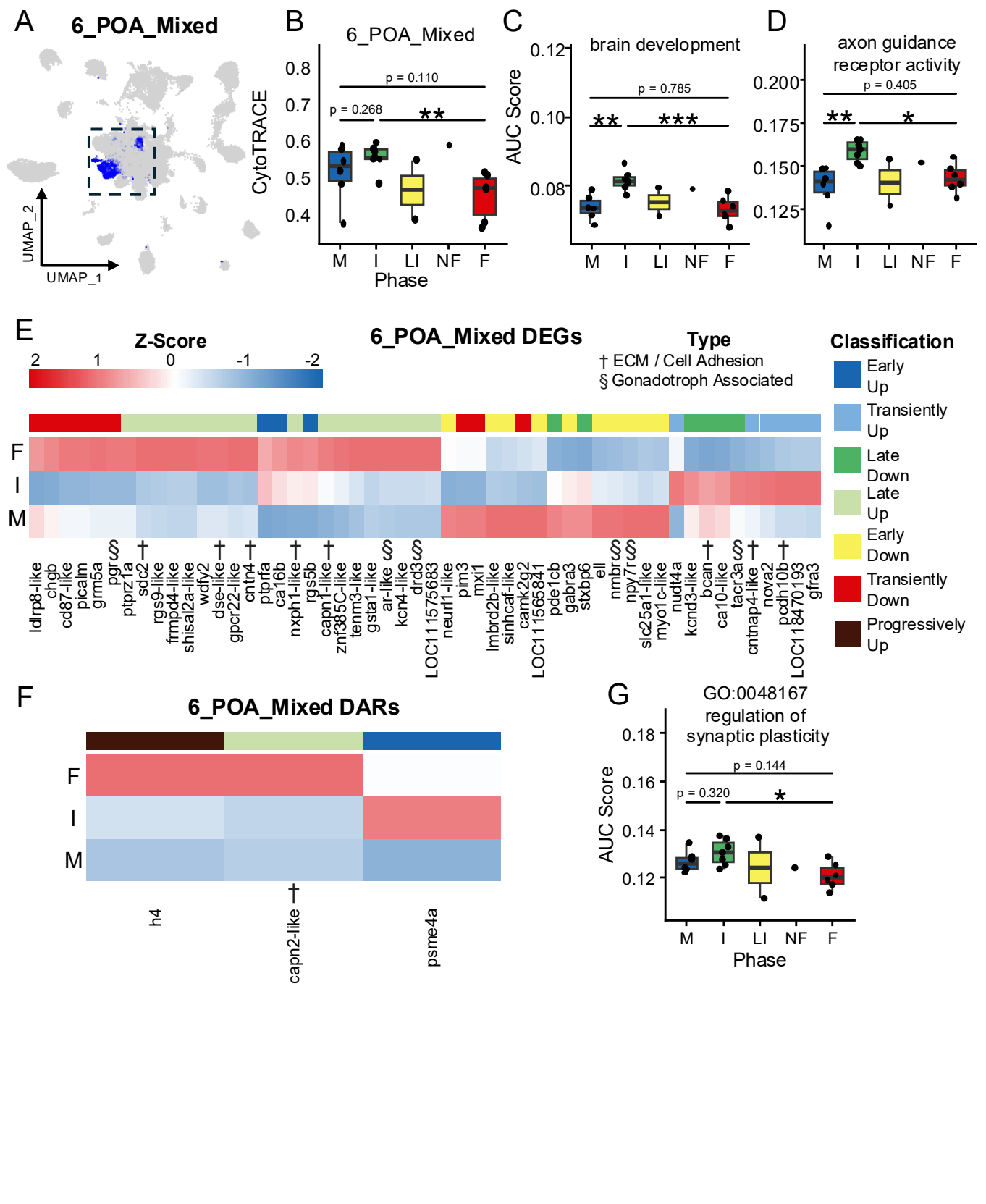

A
6_POA_Mixed
UMAP_2
UMAP_1
B
6_POA_Mixed
CytoTRACE
p = 0.110
p = 0.268
**
M I LI NF F
Phase
C
brain development
AUC Score
p = 0.785
**
***
M I LI NF F
D
axon guidance receptor activity
p = 0.405
**
*
M I LI NF F
E
Z-Score
2 1 0 -1 -2
6_POA_Mixed DEGs
Type
† ECM / Cell Adhesion
§ Gonadotroph Associated
Classification
Early Up
Transiently Up
Late Down
Late Up
Early Down
Transiently Down
Progressively Up
F
I
M
ldlrp8-like
chgb
cd87-like
picalm
grm5a
pgrc§
ptprz1a
sdc2†
rgs9-like
frmpd4-like
shisal2a-like
wdfy2
dse-like†
gpcr22-like†
cntn4
ptprfa
ca16b
nxph1-like†
rgs5b
capn1-like†
znf385C-like
tenm3-like
gsta1-like
ar-like§
kcn4-like
drd3§
LOC111575683
neurl1-like
pim3
mxi1
lmbrd2b-like
sinhcaf-like
camk2g2
LOC111565841
pde1cb
gabra3
stxbp6
ell
nmbr§§
npy7r§§
slc25a1-like
myo1c-like
nudt4a
kcnd3-like†
bcan
ca10-like
§†
tacr3a
cntnap4-like
nova2†
pcdh10b
LOC118470193
gfra3
F
6_POA_Mixed DARs
F
I
M
h4
capn2-like†
psme4a
G
GO:0048167 regulation of synaptic plasticity
AUC Score
p = 0.144
p = 0.320
*
M I LI NF F
Phase

Fig.5. 6_POA_Mixed undergoes changes in transcriptome and epigenome consistent with neurogenesis.
(A) UMAP projection with 6_POA_Mixed highlighted.
(B) CytoTRACE score for 6_POA_Mixed as a whole.
(C) AUCell score for a gene expression module made of genes annotated with the GO term “brain development”
(D) AUCell score for a gene expression module made of genes annotated with the GO term “axon guidance receptor activity”.
(E) DEGs in 6_POA_Mixed. In B, C, cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed.
(F) DARs in 6_POA_mixed..
(G)Prortion of cells in 6_POA_Mixed that are ECM DEG+
In B-D, G, points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

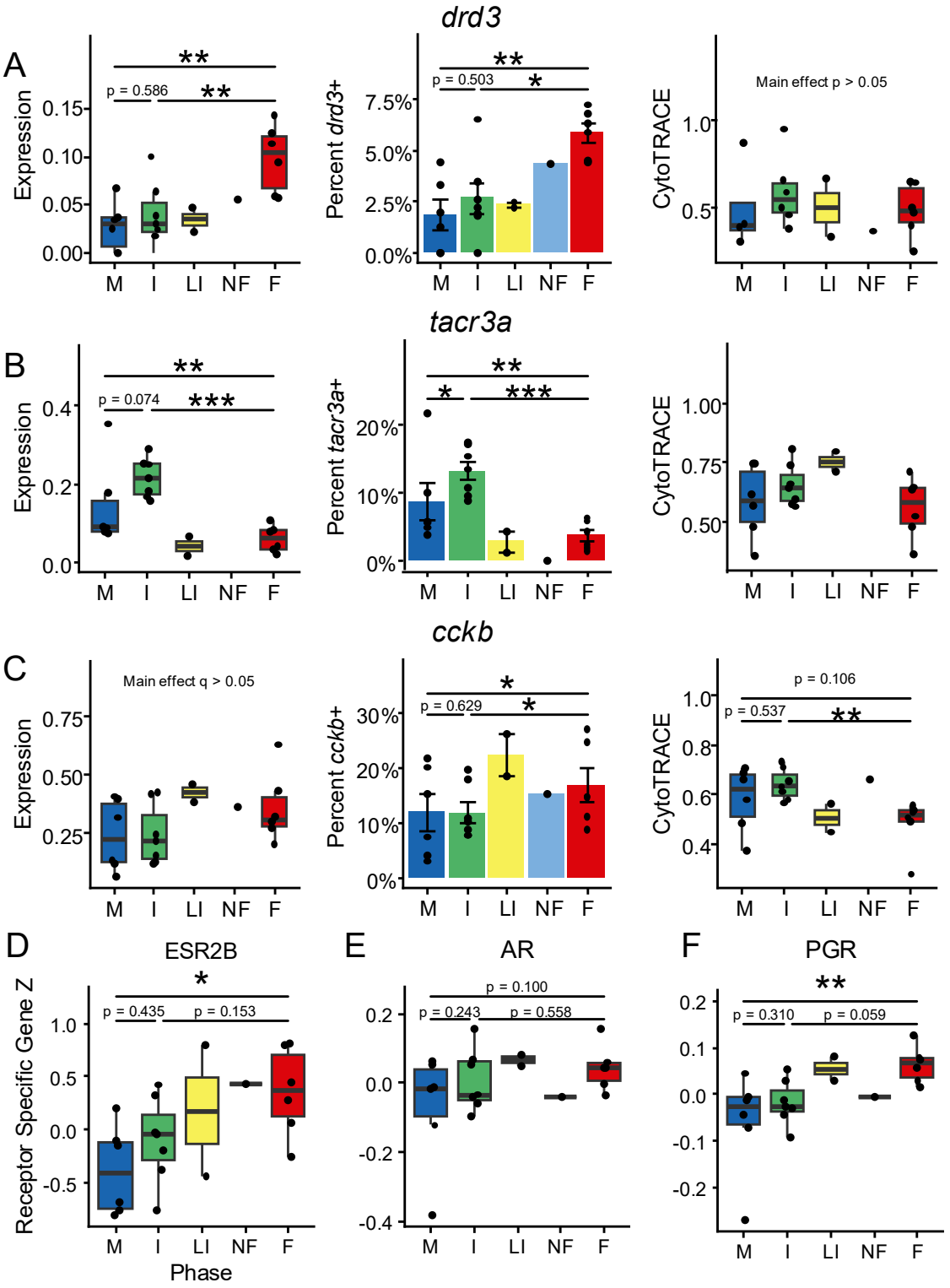

A
drd3
Expression
Percent drd3+
CytoTRACE
p = 0.586
**
p = 0.503
*
Main effect p > 0.05
B
tacr3a
Expression
Percent tacr3a+
CytoTRACE
p = 0.074
***
C
cckb
Expression
Percent cckb+
CytoTRACE
Main effect q > 0.05
p = 0.629
p = 0.537
p = 0.106
D
ESR2B
Receptor Specific Gene Z
p = 0.435
p = 0.153
E
AR
p = 0.243
p = 0.558
p = 0.100
F
PGR
p = 0.310
p = 0.059
M
I
LI
NF
F
Phase

Fig.6. Changes to 6_POA_Mixed Drive Differential Regulation of Gonadotrophs.
(A) Left panel expression of *drd3* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *drd3* by phase. Right panel: CyoTRACE score for *drd3*+ cells by phase.
(B) Left panel expression of *tacr3a* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *tacr3a* by phase. Right panel: CyoTRACE score for *tacr3a*+ cells by phase.
(C) Left panel expression of *cckb* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *cckb* by phase. Right panel: CyoTRACE score for *cckb* + cells by phase.
(D) xx justin can you describe what these are
(E) xx justin can you describe what these are
(F) xx justin can you describe what these are
Points indicate the mean for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$. "Main effect Main effect $p > 0.05$" indicates type III tests were not significant so pairwise comparisons were not computed.

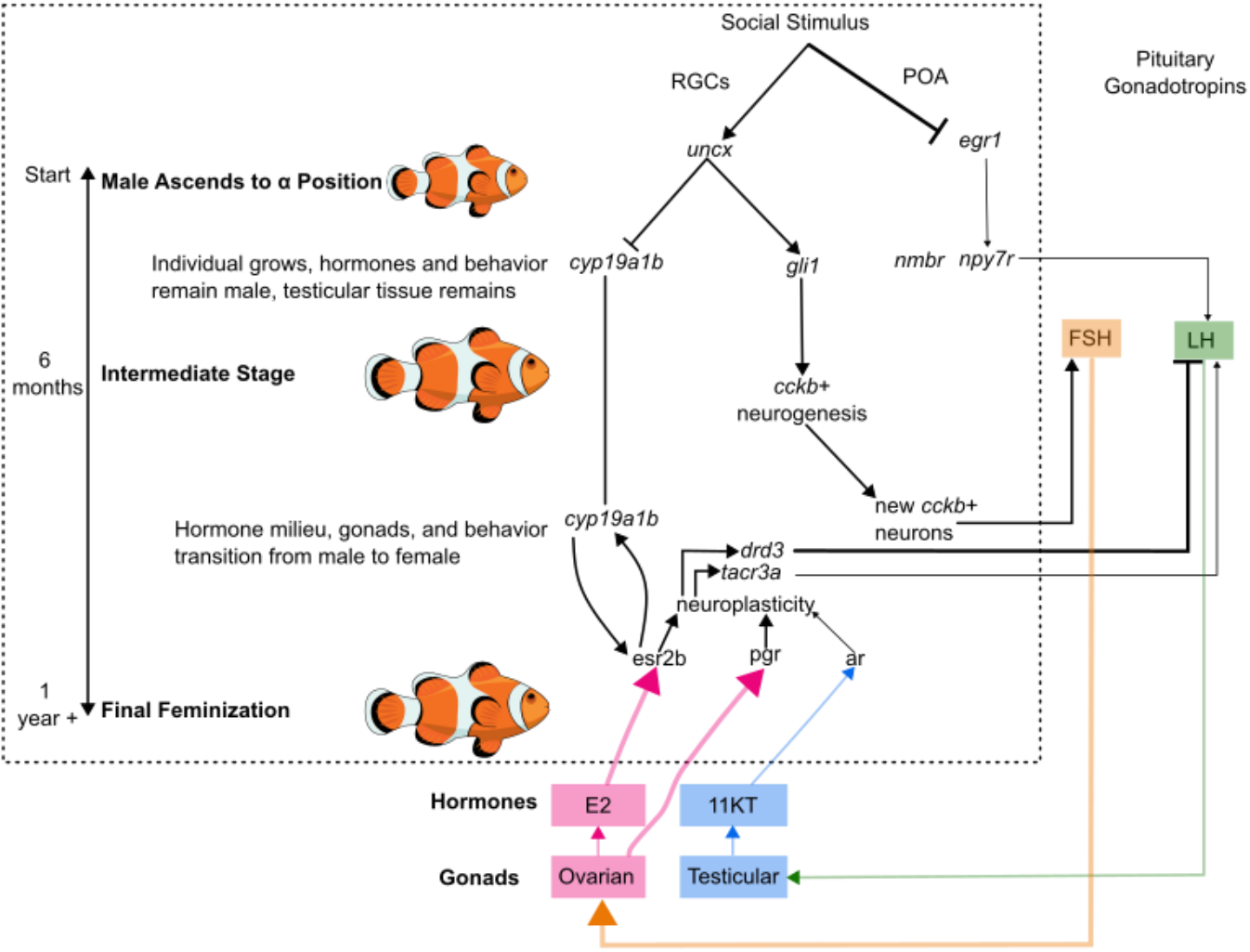
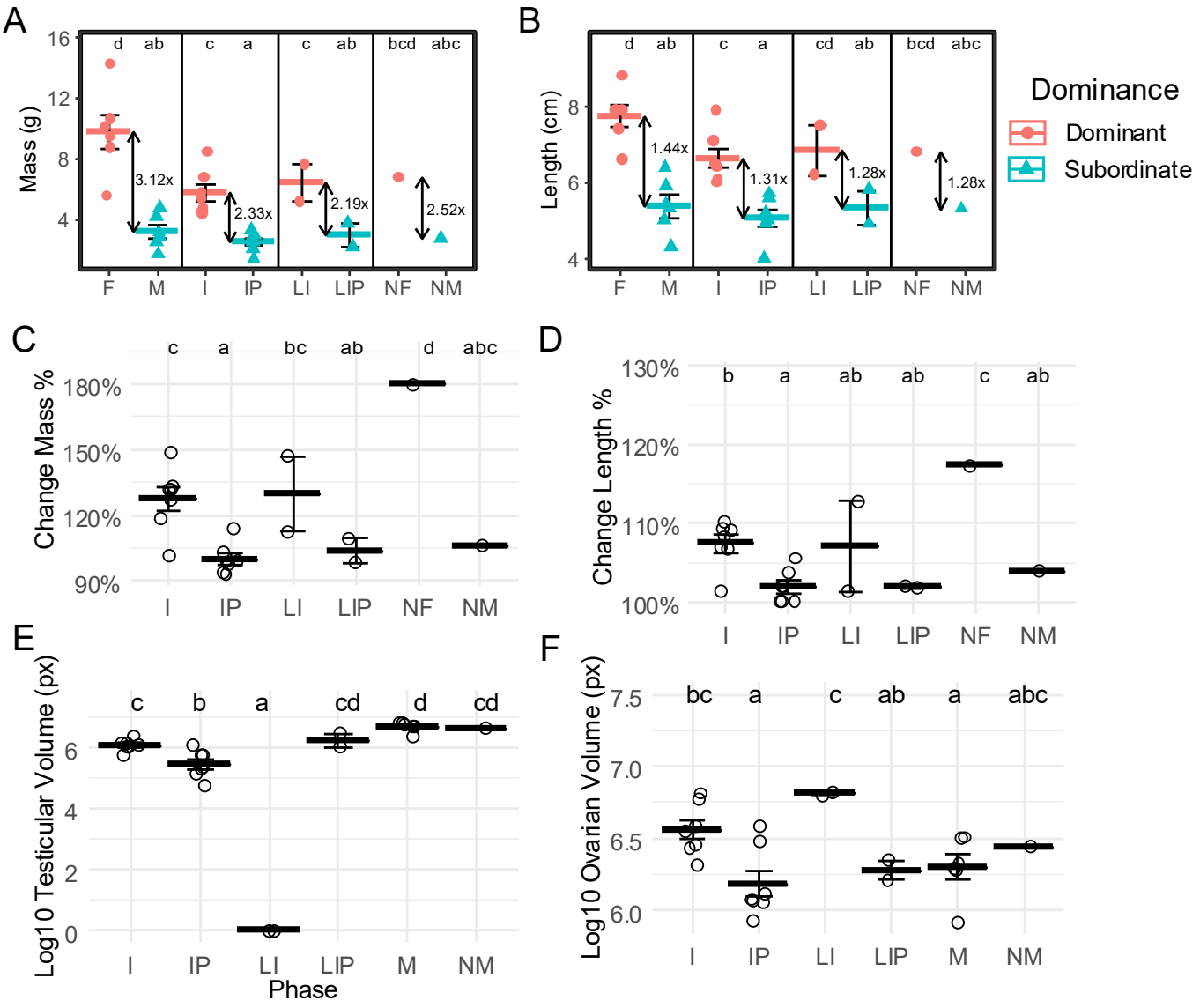
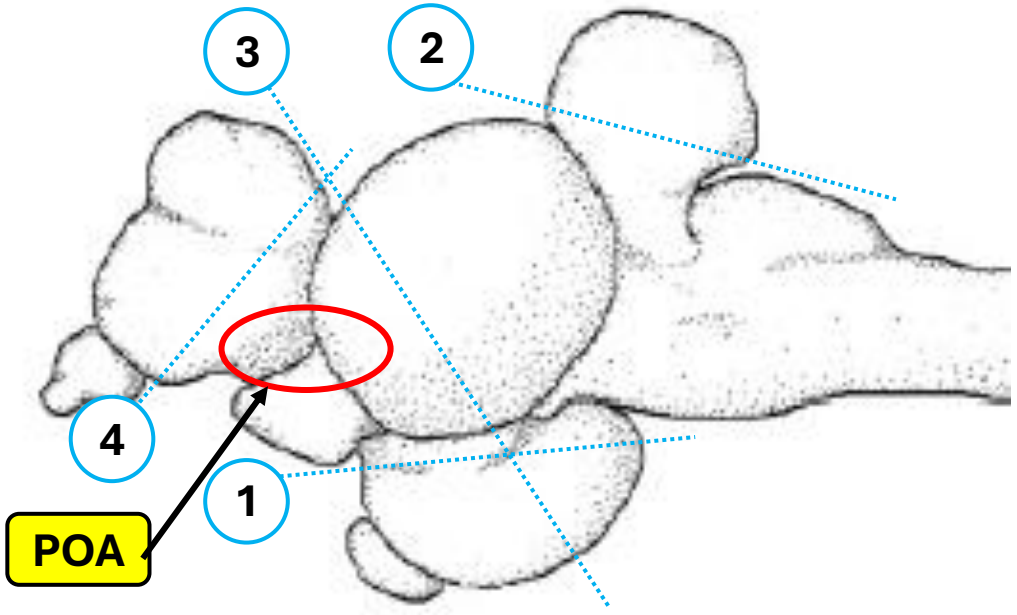


Fig.7 Schematic outlining our model of the changes that occur during protandours sex change in *A. ocellaris.* Thicker lines indicate increased output, thinner lines indicate decreased output. Arrows indicate stimulation and blunt ends indicate inhibition.



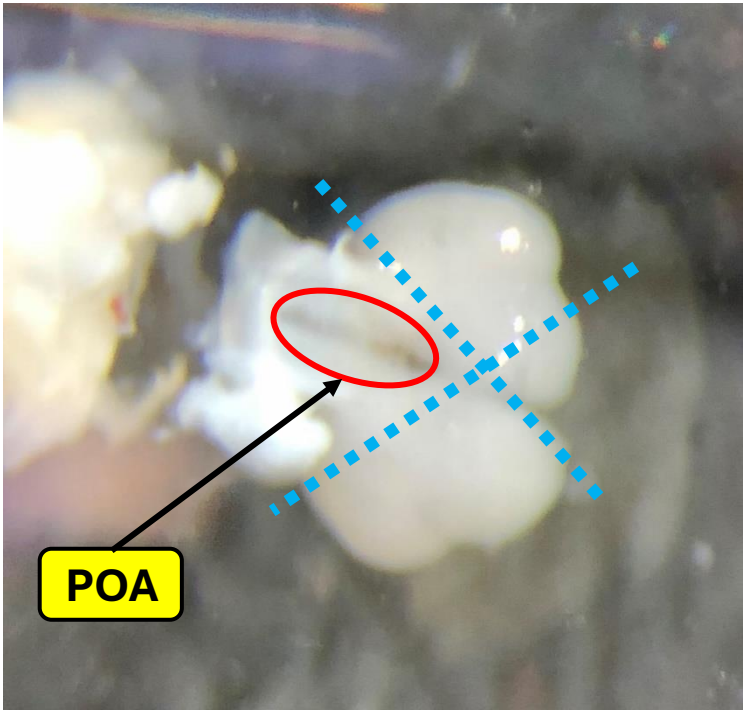
Supplementary Fig.1.
(A) Body mass by phase.
(B) Body length by phase. Plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.
(C) Change in body mass by phase.
(D) Change in length by phase.
(E) Log10 testicular volume estimate by phase.
(F) Log10 ovarian volume estimate by phase.
Crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different. In A, B Arrows indicate fold-difference between the two groups.

A



Blue dotted lines represent cuts made

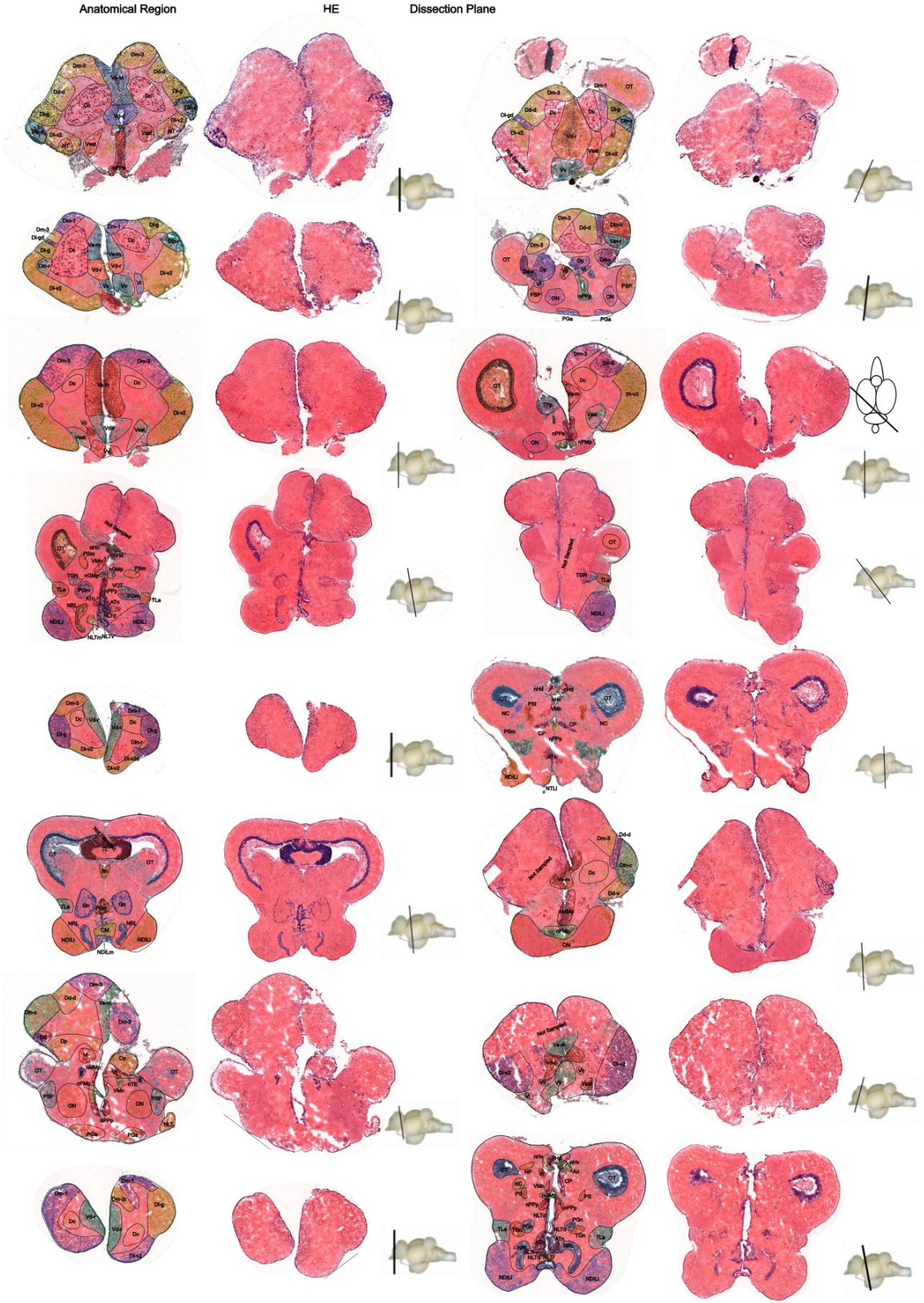
B



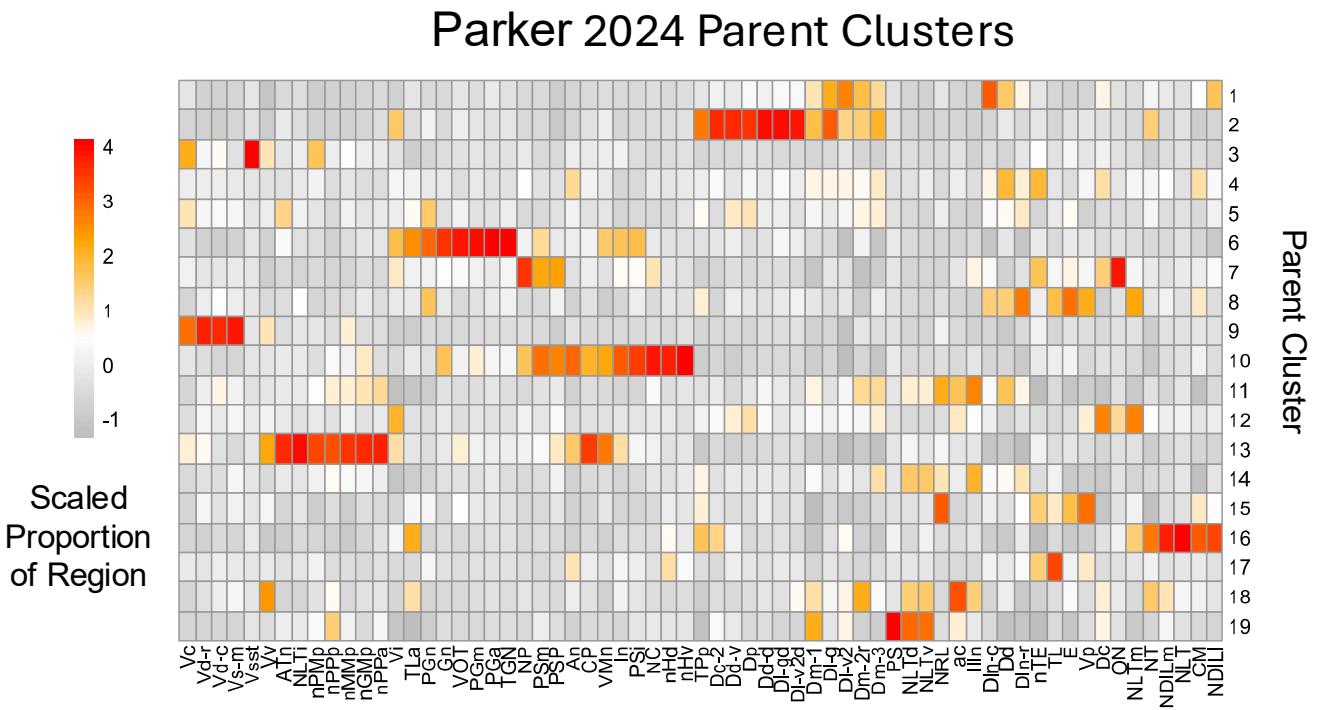
Supplementary Fig.2.

(A) Side view of anemonefish brain with dissections indicated by dashes blue lines. The number indicates the order of cuts.

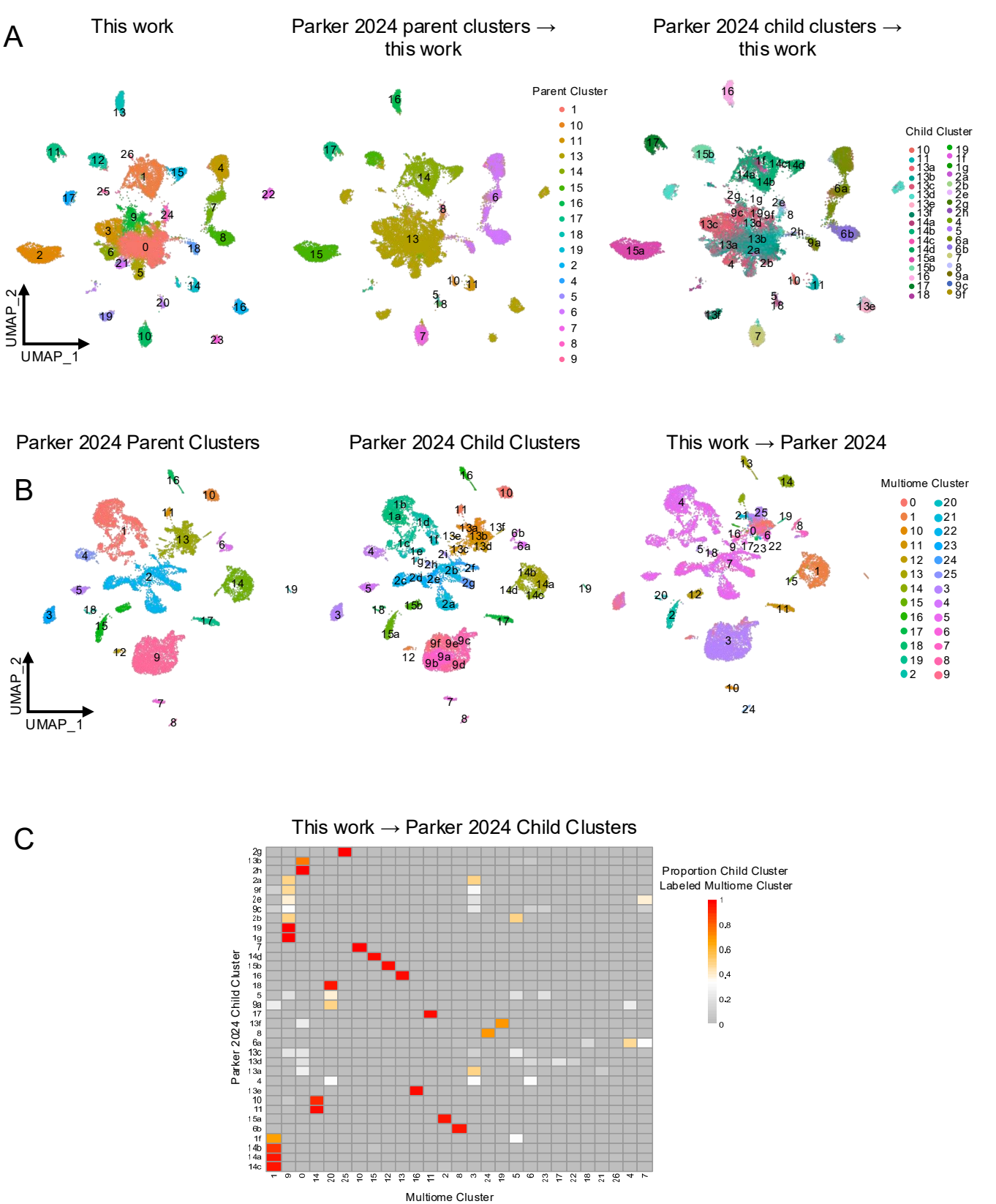
(B) Top down vies of anemonefish brain with dissections indicated by dashed blue lines. In A and B, the POA is highlighted with a red circle


Anatomical Region
HE
Dissection Plane

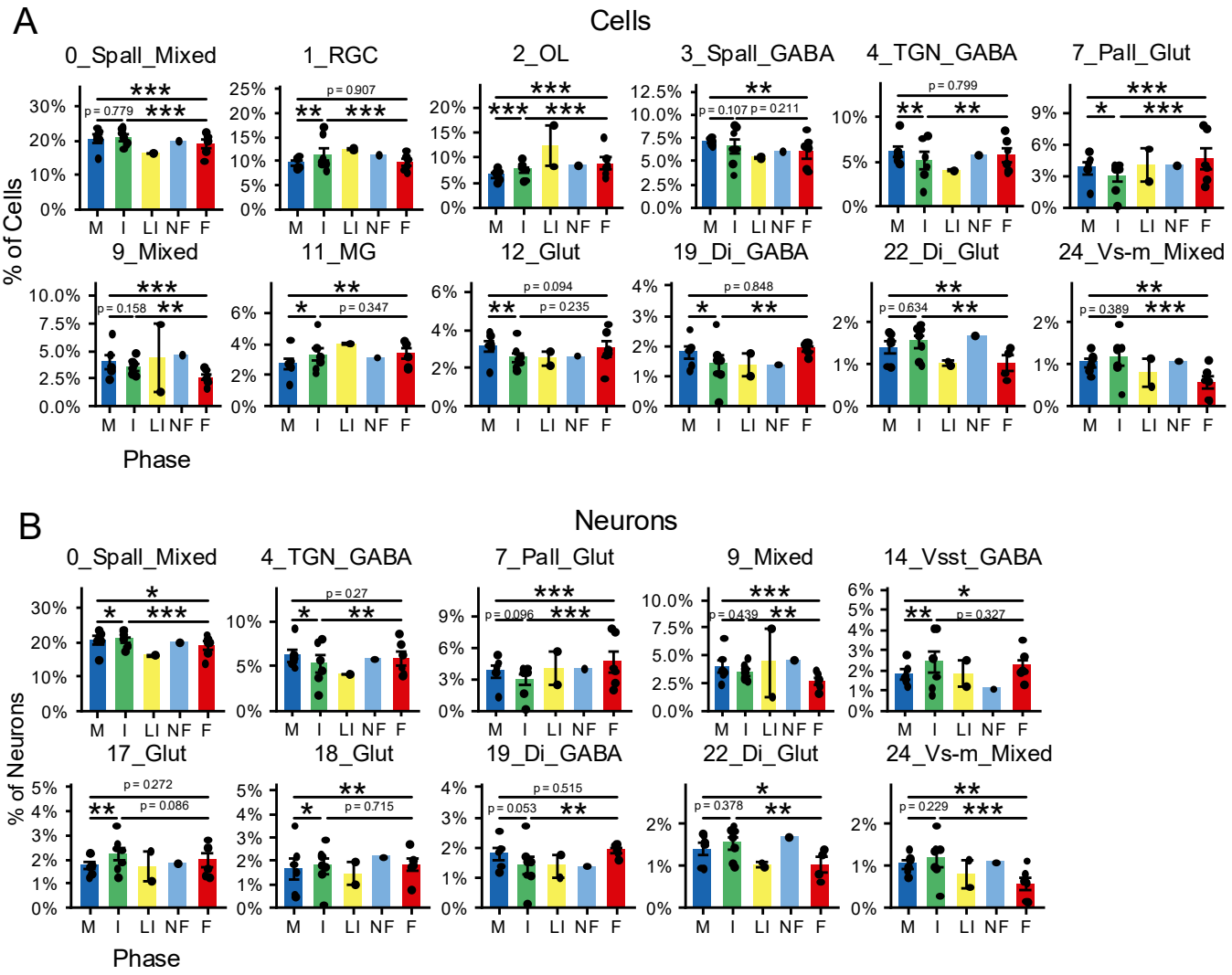
Supplementary Fig.3. Drawings of each section sequenced with Visium HD with neuroanatomical regions labeled. The middle column displays the unlabeled hematoxylin and eosin stain and the right column shows a small brain from the side with a black line indicating the sectioning plane for the given section.



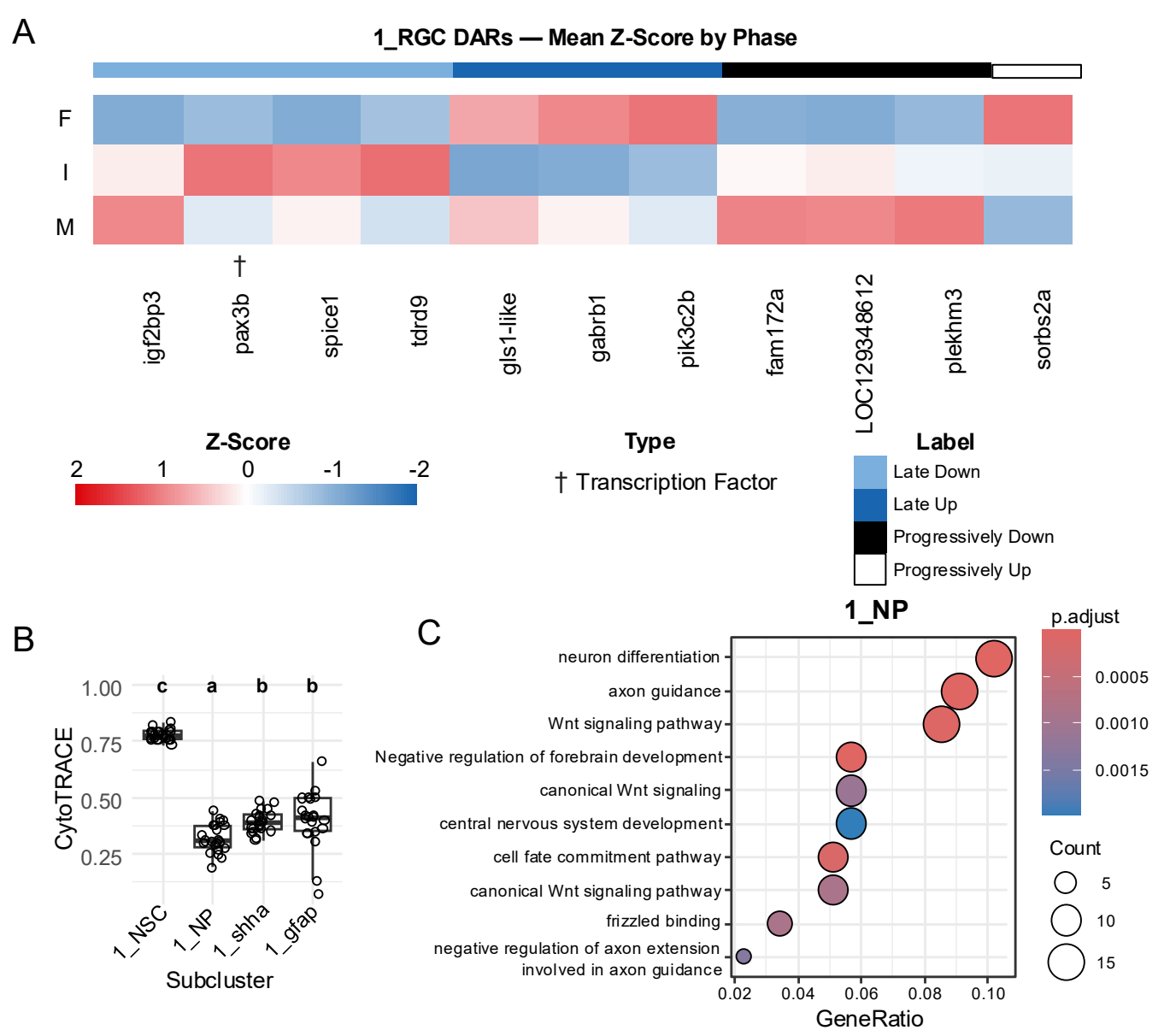
Supplementary Fig.4 Heatmap of Parker 2024 parent clusters mapped onto the Visium HD data in this work. The color in the heatmap indicates the scaled proportion of cells from each parent cluster composing each neuroanatomical region.


A
This work
Parker 2024 parent clusters → this work
Parker 2024 child clusters → this work
Parent Cluster
Child Cluster
UMAP_2
UMAP_1
B
Parker 2024 Parent Clusters
Parker 2024 Child Clusters
This work → Parker 2024
Multiome Cluster
C
This work → Parker 2024 Child Clusters
Parker 2024 Child Cluster
Multiome Cluster
Proportion Child Cluster Labeled Multiome Cluster

Supplementary Fig.5 Results of label transfer from Parker 2024 dataset.
(A) Projection of Parker 2024 parent and child clusters onto the current work.
(B) Clusters from the current work projected onto the Parker 2024 object.
(C) Heatmap showing the proportion of each child cluster in the Parker 2024 labeled with the clusters from this work in a multiome to Parker 2024 child cluster mapping.



Supplementary Fig.6.
(A) Percent of all cells belonging to each multiome cluster.
(B) Precent of all neurons belonging to each multiome cluster.
Only significant results are shown. Points indicate the mean for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$.


A
1_RGC DARs — Mean Z-Score by Phase
F
I
M
†
igf2bp3
pax3b
spice1
tdrd9
gls1-like
gabrb1
pik3c2b
fam172a
LOC129348612
plekhm3
sorbs2a
Z-Score
2
1
0
-1
-2
Type
† Transcription Factor
Label
Late Down
Late Up
Progressively Down
Progressively Up
B
1.00
0.75
0.50
0.25
CytoTRACE
c
a
b
b
1_NSC
1_NP
1_shha
1_gfap
Subcluster
C
1_NP
neuron differentiation
axon guidance
Wnt signaling pathway
Negative regulation of forebrain development
canonical Wnt signaling
central nervous system development
cell fate commitment pathway
canonical Wnt signaling pathway
frizzled binding
negative regulation of axon extension involved in axon guidance
0.02
0.04
0.06
0.08
0.10
GeneRatio
p.adjust
0.0005
0.0010
0.0015
Count
5
10
15

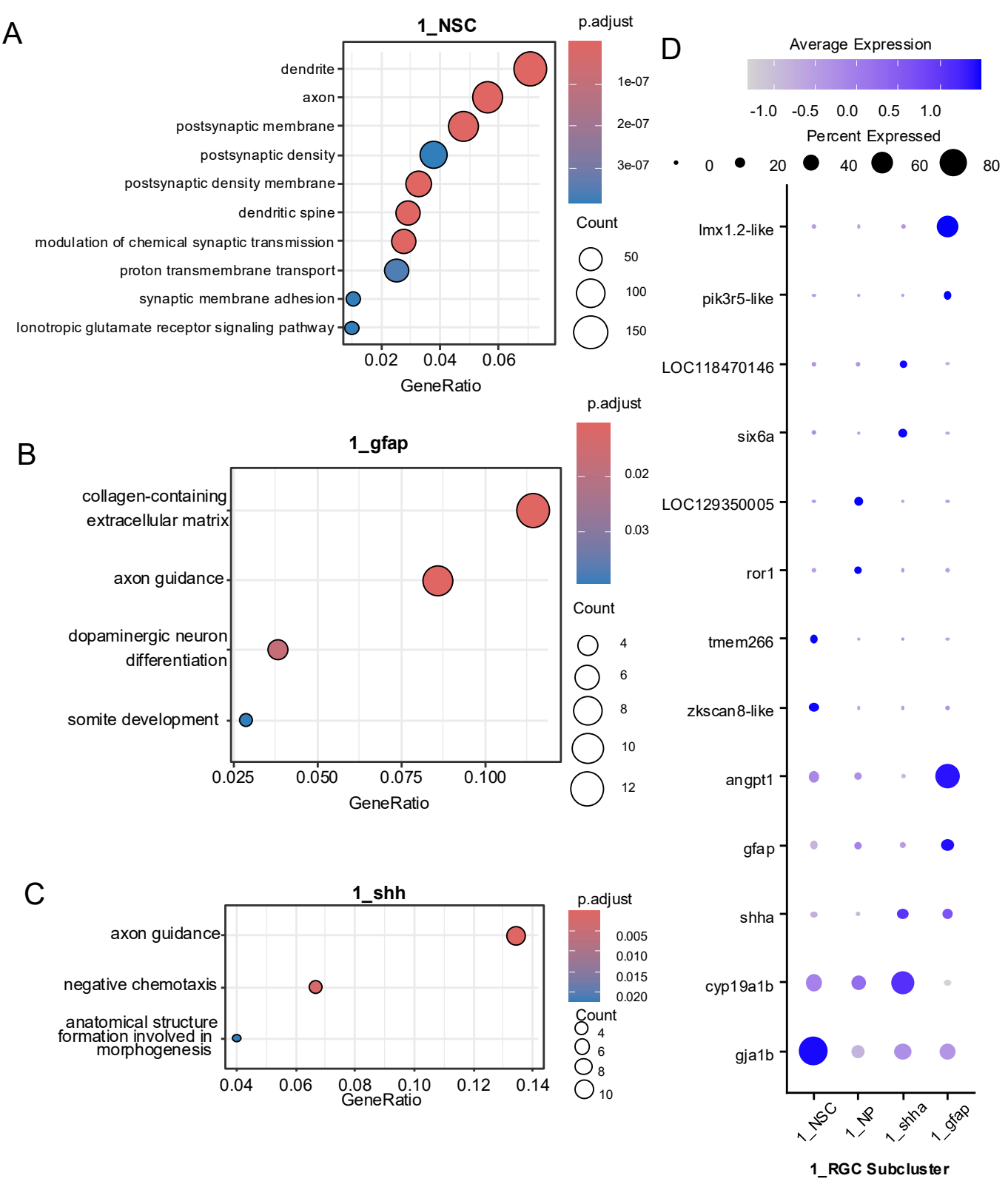
Supplementary Fig.7.

(A) DARs in 1_RGC. Cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed.

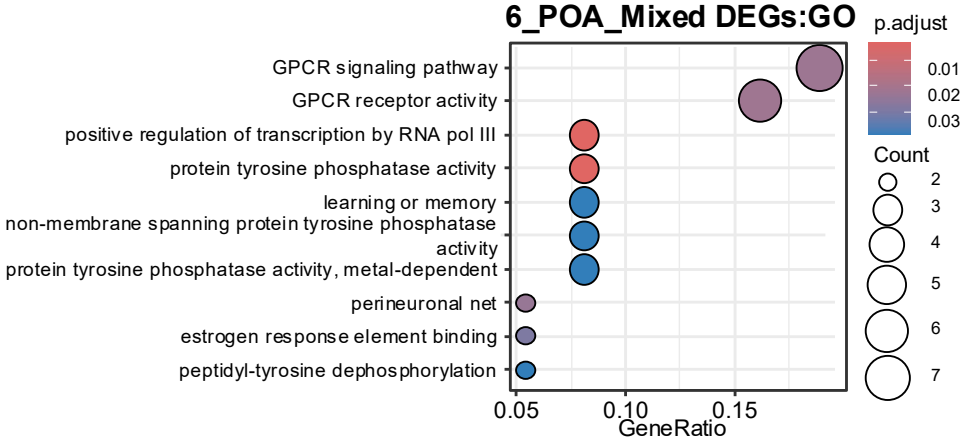
(B) CytoTRACE score by 1_RGC subcluster. Letters above indicate significance, if two subclusters share a letter they are not significantly different.

(C) Gene ontology analysis of marker genes for 1_NP.

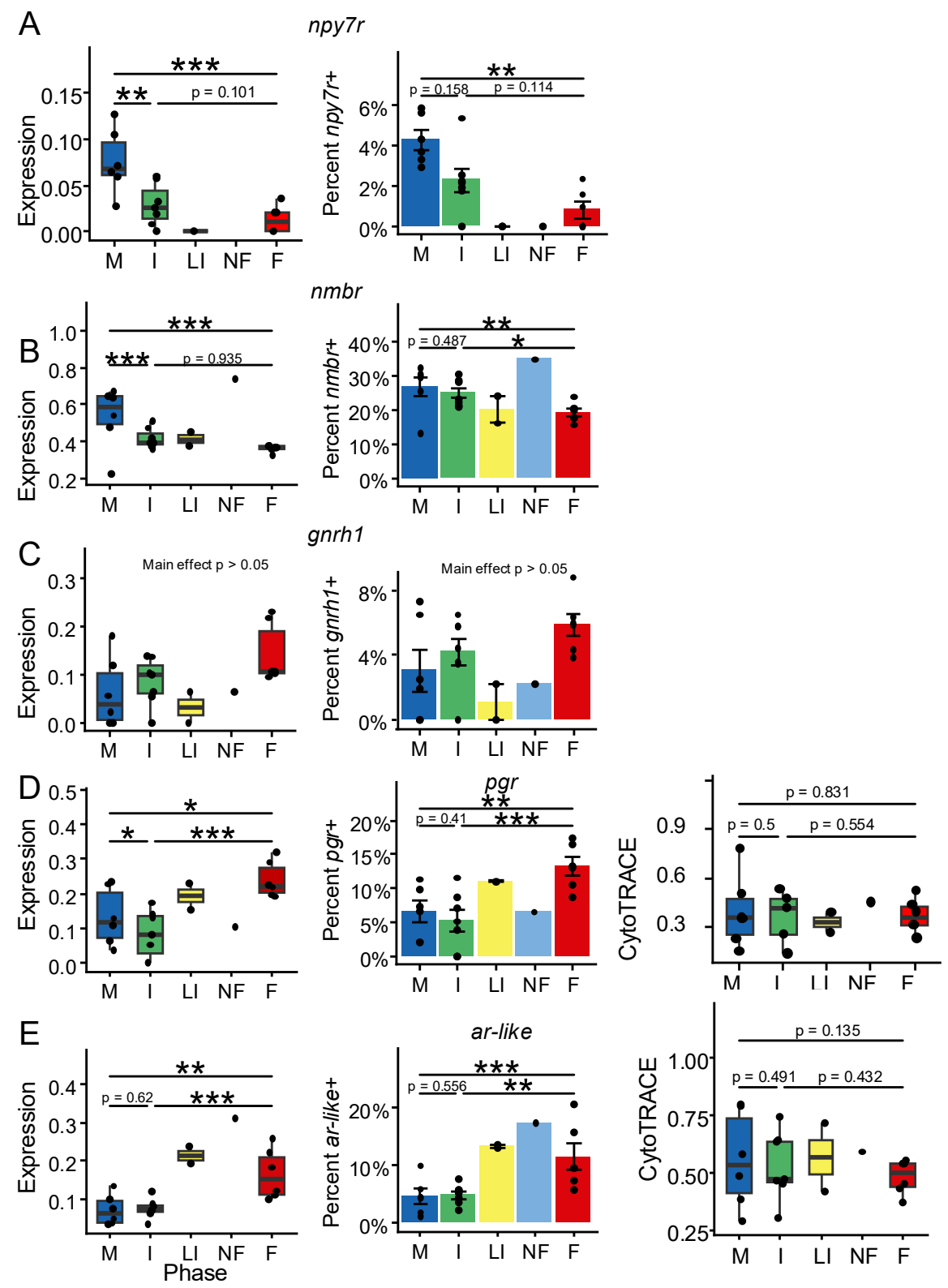
Color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list.


A
1_NSC
dendrite
axon
postsynaptic membrane
postsynaptic density
postsynaptic density membrane
dendritic spine
modulation of chemical synaptic transmission
proton transmembrane transport
synaptic membrane adhesion
Ionotropic glutamate receptor signaling pathway
0.02
0.04
0.06
GeneRatio
p.adjust
1e-07
2e-07
3e-07
Count
50
100
150
B
1_gfap
collagen-containing extracellular matrix
axon guidance
dopaminergic neuron differentiation
somite development
0.025
0.050
0.075
0.100
GeneRatio
p.adjust
0.02
0.03
Count
4
6
8
10
12
C
1_shh
axon guidance
negative chemotaxis
anatomical structure formation involved in morphogenesis
0.04
0.06
0.08
0.10
0.12
0.14
GeneRatio
p.adjust
0.005
0.010
0.015
0.020
Count
4
6
8
10
D
Average Expression
-1.0
-0.5
0.0
0.5
1.0
Percent Expressed
0
20
40
60
80
lmx1.2-like
pik3r5-like
LOC118470146
six6a
LOC129350005
ror1
tmem266
zkscan8-like
angpt1
gfap
shha
cyp19a1b
gja1b
1_NSC
1_NP
1_shha
1_gfap
1_RGC Subcluster

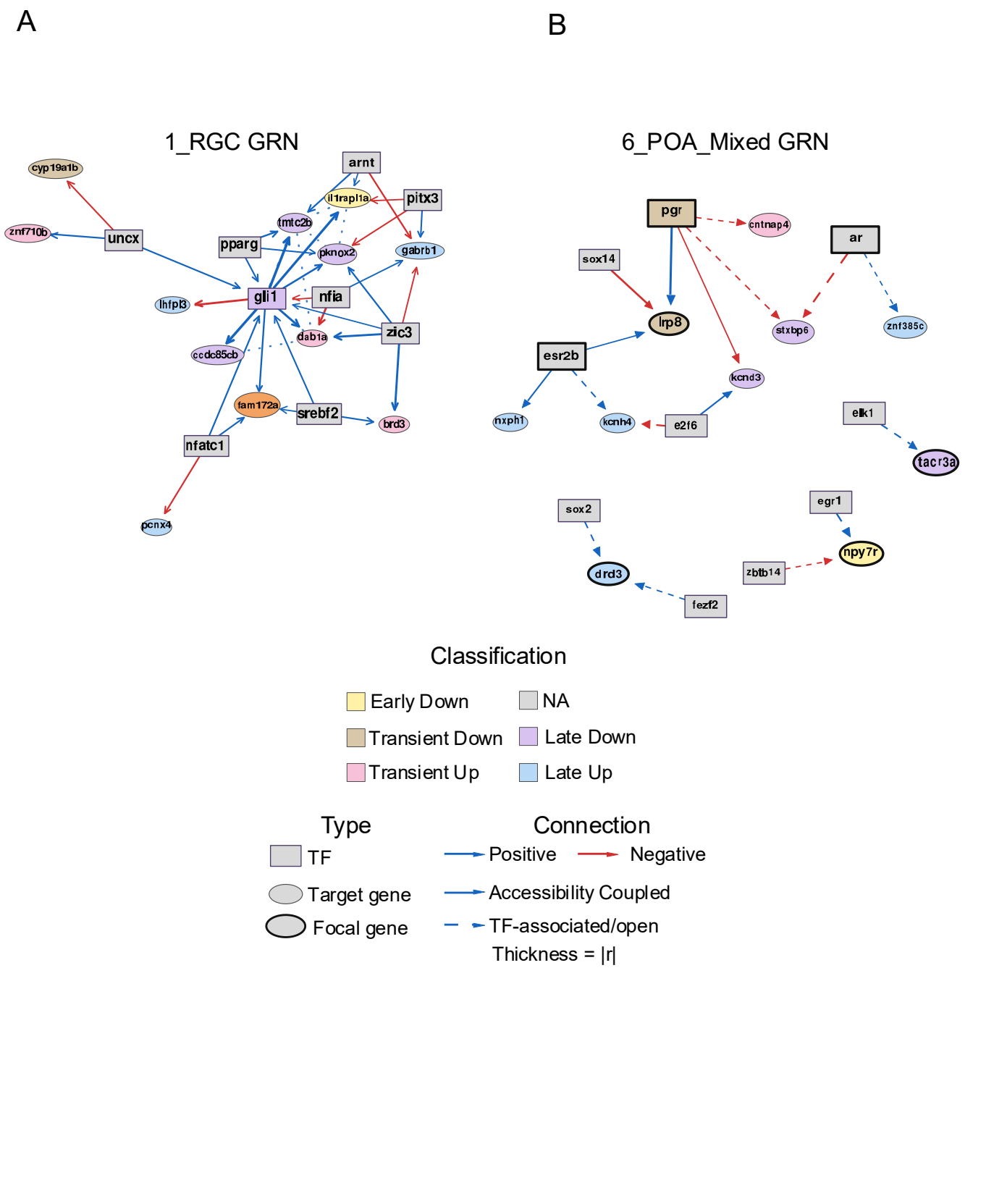
Supplementary Fig.8. Gene ontology (GO) analysis for the marker genes of
(A) 1_NSC,
(B) 1_gfap, and
(C) 1_shha.
In A-C, color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list. (D) Canonical and data-driven marker genes for each 1_RGC subcluster



Supplementary Fig.9. GO analysis of all of the DEGs in 6_POA_Mixed. Ccolor indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list.


A
npy7r
Expression
Percent npy7r+
p = 0.101
p = 0.158
p = 0.114
B
nmbr
Percent nmbr+
p = 0.935
p = 0.487
C
gnrh1
Main effect p > 0.05
Percent gnrh1+
D
pgr
Percent pgr+
p = 0.41
CytoTRACE
p = 0.831
p = 0.5
p = 0.554
E
ar-like
Percent ar-like+
p = 0.62
p = 0.556
p = 0.135
p = 0.491
p = 0.432
M
I
LI
NF
F
Phase

Supplementary Fig.10. Expression (left) and percent of 6_POA_Mixed (right) expressing
(A) npy7r,
(B) *nmbr*,
(C) *gnrh1.* Expression (left) and percent of 6_POA_Mixed (middle), and CytoTRACE score (right) of cells expressing
(D) *pgr*,
(E), *ar-like.*
Points indicate the mean for each individual. Asterisks: "*": $p < 0.05$, "**": $p < 0.01$, "***": $p < 0.001$.


A
B
1_RGC GRN
6_POA_Mixed GRN
arnt
cyp19a1b
il1rapl1a
pitx3
tmtc2b
znf710b
uncx
pparg
pknox2
gabrb1
lhfpl3
gli1
nfia
dab1a
zic3
ccdc85cb
fam172a
srebf2
brd3
nfatc1
pcnx4
pgr
cntnap4
ar
sox14
lrp8
stxbp6
znf385c
esr2b
kcnd3
nxph1
kcnh4
e2f6
elk1
tacr3a
sox2
egr1
npy7r
drd3
zbtb14
fezf2
Classification
Early Down
NA
Transient Down
Late Down
Transient Up
Late Up
Type
TF
Target gene
Focal gene
Connection
Positive
Negative
Accessibility Coupled
TF-associated/open
Thickness = |r|

Supplementary Fig. 11 xx Justin please write